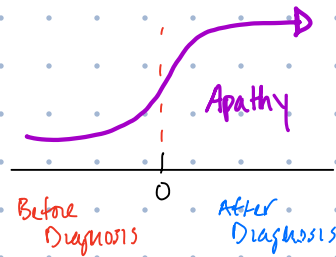


↑ Antidepressant medications



How to adjust for disease severity?

We can use a proxy that carries similar info as U

$$R = \frac{\text{Age at disease onset}}{\text{Current Age}}$$

Example 1: Age at disease onset: 25
Age at treatment initiation: 50

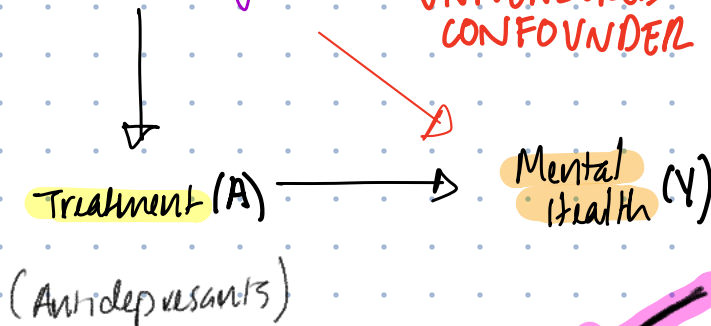
$$\Rightarrow R = \frac{25}{50} = 0.5$$

Example 2: Age at disease onset: 10
Age at treatment initiation: 50

$$\Rightarrow R = \frac{10}{50} = 0.2$$

Disease severity (U)

UNMEASURED CONFOUNDER



— STATISTICAL PROBLEM —

A = Treatment (0,1)
Y = Mental Health (1 year)
Z = Age at treatment initiation
X = Age at disease onset

However, instead of observing X, we observe (W, Δ) such that $W = \min(X, c)$ $\Delta = I(X \leq c)$

Important because we need

$$R = \frac{\text{Age at disease onset}}{\text{Current Age}} = \frac{X}{Z}$$

but we have a bounded $\frac{W}{Z}$

After many assumptions:

More information:

